

GoRest REST API Testing Project

Software Test Plan (IEEE Style)

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Project	GoRest REST API Testing Project
Date	11 September 2026
Base URL	https://gorest.co.in/public/v2

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1. Introduction

This Test Plan describes the testing strategy for the GoRest REST API Testing Project. The project focuses on validating the GoRest Public API using Postman. The testing covers CRUD operations, response validation, environment variable chaining and negative testing scenarios. The purpose of this document is to explain what will be tested, how the testing will be performed and what documents will be produced after testing.

2. Project Overview

The GoRest Public API provides user management endpoints that allow creating, viewing, updating, replacing and deleting users. This project tests those endpoints using Postman and verifies whether the API behaves as expected. The project also includes negative testing to ensure that invalid requests are handled correctly.

Tools Used

Tool	Purpose
Postman	API Testing
Newman	Collection Execution
HTML Extra Reporter	HTML Report
GoRest Public API	Test Environment
GitHub	Portfolio

3. Objectives

The main objectives of this testing are: Verify CRUD operations, validate API responses, check authentication, verify response headers, validate returned data, test error handling and generate a professional Newman report.

4. Scope of Testing

In Scope

- Create User
- Fetch User
- List Users
- Update User (PATCH)
- Verify PATCH Update
- Replace User (PUT)
- Verify PUT Update
- Delete User
- Verify Deleted User
- Invalid Token
- Invalid User ID
- Create Without Email
- Duplicate Email
- Invalid Gender
- Empty Name
- Missing Status
- Wrong Endpoint
- Wrong Method
- Status Codes
- Content-Type
- Environment Variables
- Required Fields
- Error Responses

5. Test Items

Endpoint	Method
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/public/v2/users	POST
/public/v2/users	GET
/public/v2/users/{id}	GET
/public/v2/users/{id}	PATCH
/public/v2/users/{id}	PUT
/public/v2/users/{id}	DELETE

6. Features to be Tested

Each feature is verified using Postman requests. Create User: Valid user creation, response validation and environment variable storage. Fetch User: User retrieval and data verification. List Users: User list availability and JSON response validation. Update User: Partial update using PATCH and verification through GET. Replace User: Complete replacement using PUT and verification through GET. Delete User: Successful deletion and verification using GET. Negative Testing: Unauthorized access, validation errors, invalid routes and invalid user IDs.

7. Features Not Tested

The following areas are outside the scope of this project: Performance Testing, Load Testing, Security Penetration Testing, Database Validation, UI Testing and Mobile Testing. These areas require additional tools and environments.

8. Test Approach

Step 1: Environment Setup

- Configure Base URL.
- Configure Bearer Token.
- Configure Environment Variables.

Step 2: CRUD Testing

1. Create User
2. Fetch User
3. List Users
4. PATCH Update
5. Verify PATCH
6. PUT Replace
7. Verify PUT
8. Delete User
9. Verify Delete

Step 3: Response Validation

- Status Code
- Content-Type
- Returned Data
- Environment Variables

Step 4: Negative Testing

- Wrong Token

- Duplicate Email
- Invalid Gender
- Wrong Endpoint

9. Test Environment

Item	Value
API	GoRest Public API
Base URL	https://gorest.co.in/public/v2
Tool	Postman
Reporter	Newman HTML Extra
Operating System	Windows
Internet	Required

10. Entry Criteria

Testing can begin when Postman is installed, internet connection is available, Bearer Token is configured, Environment Variables are ready and the Collection is imported.

11. Exit Criteria

Testing is considered complete when all CRUD requests execute successfully, negative tests execute successfully, expected status codes are verified, environment variables work correctly, the Newman report is generated and documentation is completed.

12. Test Deliverables

Deliverable	Status
Postman Collection	Completed
Environment File	Completed
Newman HTML Report	Completed
Test Cases	Completed
Bug Report	Completed
Test Plan	Completed
Test Summary Report	Pending
RTM	Pending

13. Risk and Assumptions

Risks

- Public API behavior may change.
- Token expiration may interrupt testing.
- Internet issues may affect execution.

Assumptions

- GoRest API remains available.
- Valid authentication token is provided.
- Environment variables are configured correctly.

14. Approval

Field	Value
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